# AuroraCounsel: A Multi-Agent Large Language Model Collaboration Framework for Psychological Counseling

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https://github.com/ananzhu/AuroraCounsel_open

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## Abstract

LLM counselors struggle with users' shifting emotions and lack stage-aware strategies. **AuroraCounsel** is a multi-agent framework spanning emotion sensing, reasoning and guided dialogue, with **GRPO** adapting the strategy at each counseling stage. It improves professionalism, empathy and interaction fit.

## 1 Introduction

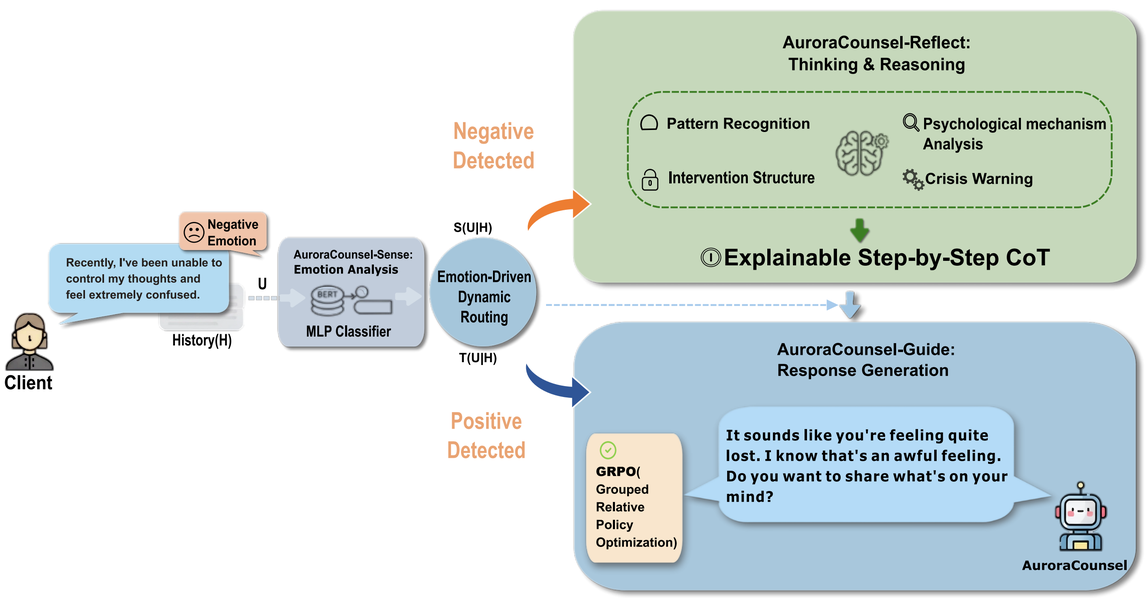
Counselors are scarce while demand keeps rising. LLMs scale, yet still lack empathy and stable reasoning.

### Our contributions

- **Sense → Reflect → Guide**: three specialised agents, one session.
- **Step-by-step CoT**: PsyDTCorpus dialogues rebuilt by Qwen-Max across four clinical dimensions.
- **Six-reward GRPO**: phase-based questioning, non-Q, repetition, length, multi-Q and in-zone rewards align AC-Guide.
- **Emotion-driven routing**: reflection only on negative emotion; smoother interaction, **41% fewer tokens**.

## 2 AuroraCounsel Collection

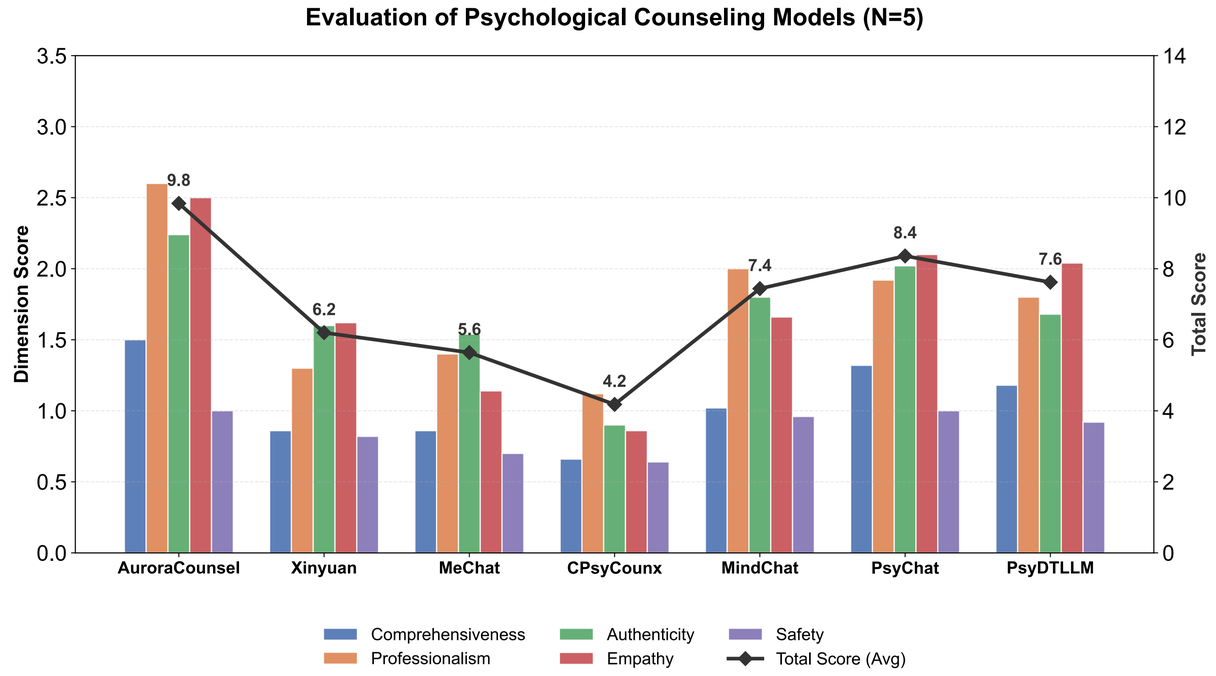
**AC-Sense** detects the client emotion (BERT + MLP, **88.0%** on 17,000 dialogues); **AC-Reflect** (Qwen2.5-7B) reasons over needs, barriers, resources and actions; **AC-Guide** replies with REBT-aligned questioning.




## 3 Experiments

### Human Expert Evaluation

Data: 240 turns/model · 5 experts




### Main results

Data: 60 dialogues / 1,100+ turns

| Type | Models | Comp. | Prof. | Auth. | Empa. | Saf. | Total |
|---|---|---|---|---|---|---|---|
| Base | Qwen2.5-7B | 1.34 ±0.07 ** | 1.92 ±0.10 *** | 1.68 ±0.12 *** | 1.54 ±0.15 *** | **1.00** ±0.00 | 7.48 ±0.37 *** |
| | Qwen3-32B | 1.47 ±0.07 | 2.28 ±0.10 *** | 1.82 ±0.13 *** | 2.05 ±0.13 *** | **1.00** ±0.00 | 8.62 ±0.35 *** |
| | Kimi-K2 | **1.56** ±0.07 | 2.49 ±0.18 ** | 1.93 ±0.18 *** | 1.84 ±0.16 *** | **1.00** ±0.00 | 8.82 ±0.51 *** |
| | DeepSeek-v3.2 | 1.51 ±0.10 | 2.42 ±0.15 *** | 2.07 ±0.13 *** | 2.09 ±0.16 *** | **1.00** ±0.00 | 9.09 ±0.43 *** |
| Domain | MeChat | 1.20 ±0.10 *** | 1.83 ±0.18 *** | 1.71 ±0.14 *** | 1.60 ±0.19 *** | **1.00** ±0.00 | 7.34 ±0.56 *** |
| | PsyChat | 1.30 ±0.13 *** | 1.84 ±0.21 *** | 1.79 ±0.17 *** | 1.91 ±0.16 *** | **1.00** ±0.00 | 7.84 ±0.59 *** |
| | PsyDTLLM | 1.20 ±0.11 *** | 2.17 ±0.14 *** | 1.69 ±0.15 *** | 1.87 ±0.18 *** | **1.00** ±0.00 | 7.93 ±0.43 *** |
| | CPsyCounX | 1.19 ±0.11 *** | 1.87 ±0.14 *** | 1.67 ±0.15 *** | 1.58 ±0.18 *** | 0.99 ±0.02 *** | 7.30 ±0.54 *** |
| | MindChat | 1.08 ±0.09 *** | 1.82 ±0.12 *** | 1.63 ±0.16 *** | 1.56 ±0.18 *** | 0.99 ±0.02 *** | 7.08 ±0.32 *** |
| | MultiAgentESC | 1.18 ±0.15 *** | 2.05 ±0.17 *** | 1.83 ±0.16 *** | 1.76 ±0.16 *** | **1.00** ±0.00 | 7.82 ±0.57 *** |
| | Xinyuan | 1.32 ±0.13 *** | 2.36 ±0.17 *** | 1.81 ±0.16 *** | 1.96 ±0.15 *** | **1.00** ±0.00 | 8.45 ±0.54 *** |
| Our | **AuroraCounsel** | 1.39 ±0.08 | **2.84** ±0.11 | **2.20** ±0.11 | **2.38** ±0.13 | **1.00** ±0.00 | **9.81** ±0.39 |

Model evaluation results across comprehensiveness (Comp.), professionalism (Prof.), authenticity (Auth.), empathy (Empa.), and safety (Saf.) (Mean ± SD). Best in bold, second-best underlined. Significance vs. AuroraCounsel (two-sided Wilcoxon signed-rank test): *p < 0.05, **p < 0.01, ***p < 0.001.

## 4 More results

- **Ablation**: routing keeps full quality (9.81) and cuts tokens by **41%**.
- **LLM-as-judge**: GPT-4o and Claude-Sonnet-4 also rank AuroraCounsel first (9.81 vs. 9.09).
- **Clinical protocol (MAGNeT)**: +11.8% collaboration, +8.8% interpersonal, +9.7% task scores.
- **Six-reward GRPO**: follows the REBT rhythm (SFT stays near 90%).
- **Safety & data**: full score (1.00), ICC(2,5) = 0.85; 2,400 expert sessions.

## Dialogue Spotlight

**我觉得如果我在一个支持和理解的环境中工作，或许会更好。但现在的我，不知道怎么才能找到那样的环境。**
**I think I might do better if I worked in a supportive and understanding environment. But right now, I have no idea how to find such a place.**

**AC-Sense:** 消极
**AC-Reflect:**
<think>
Step 1: 识别需求 →渴望支持性工作环境
Step 2: 探索阻碍 → 当前寻找环境的障碍
Step 3: 资源评估 → 可利用的支持系统
Step 4: 行动建议 → 具体可行的第一步
</think>
您希望在一个支持和理解的环境中工作，这是一个很好的愿望。目前看来，您可能感到有些迷茫，不知道如何找到这样的环境。首先，我们可以一起探讨一下，您认为目前有哪些资源可以支持您？比如，您是否考虑过加入一些行业社群，或者与已经在您感兴趣领域工作的朋友交流？
**AC-Sense:** 消极
**AC-Reflect:**
<think>
Step 1: Identify Needs → A desire for a supportive and understanding work environment
Step 2: Explore Barriers → Obstacles to finding such an environment currently
Step 3: Resource Assessment → Available support systems
Step 4: Action Recommendations → Specific and feasible first steps
</think>
It's a perfectly valid wish to hope for a supportive and understanding work environment. Right now, it seems like you might be feeling a little lost, not knowing how to find such a place. First off, we can explore together what resources you have at your disposal right now. For instance, have you considered joining any industry communities, or reaching out to friends who already work in fields you're interested in?

AuroraCounsel

**Ethical note:** AuroraCounsel is a research prototype for supportive counseling dialogue. Its output is not a diagnosis and must never replace professional medical decision-making.